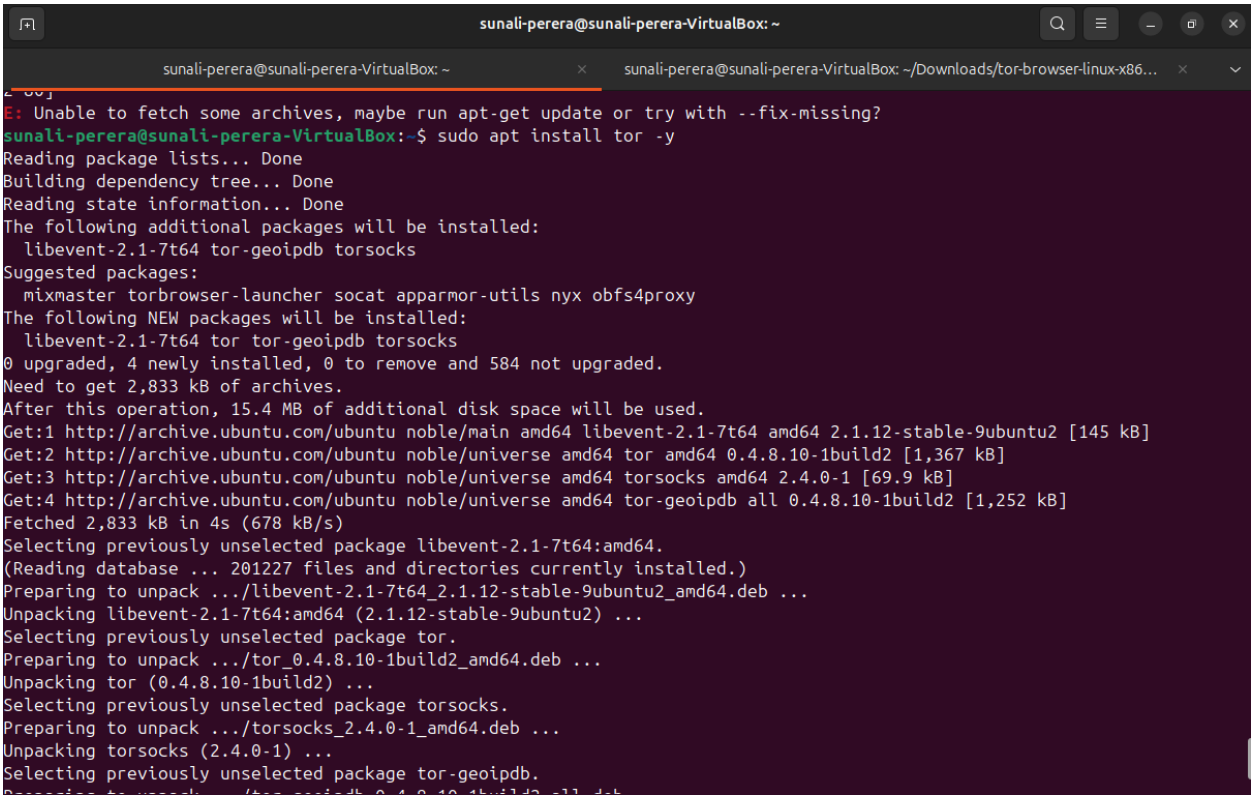
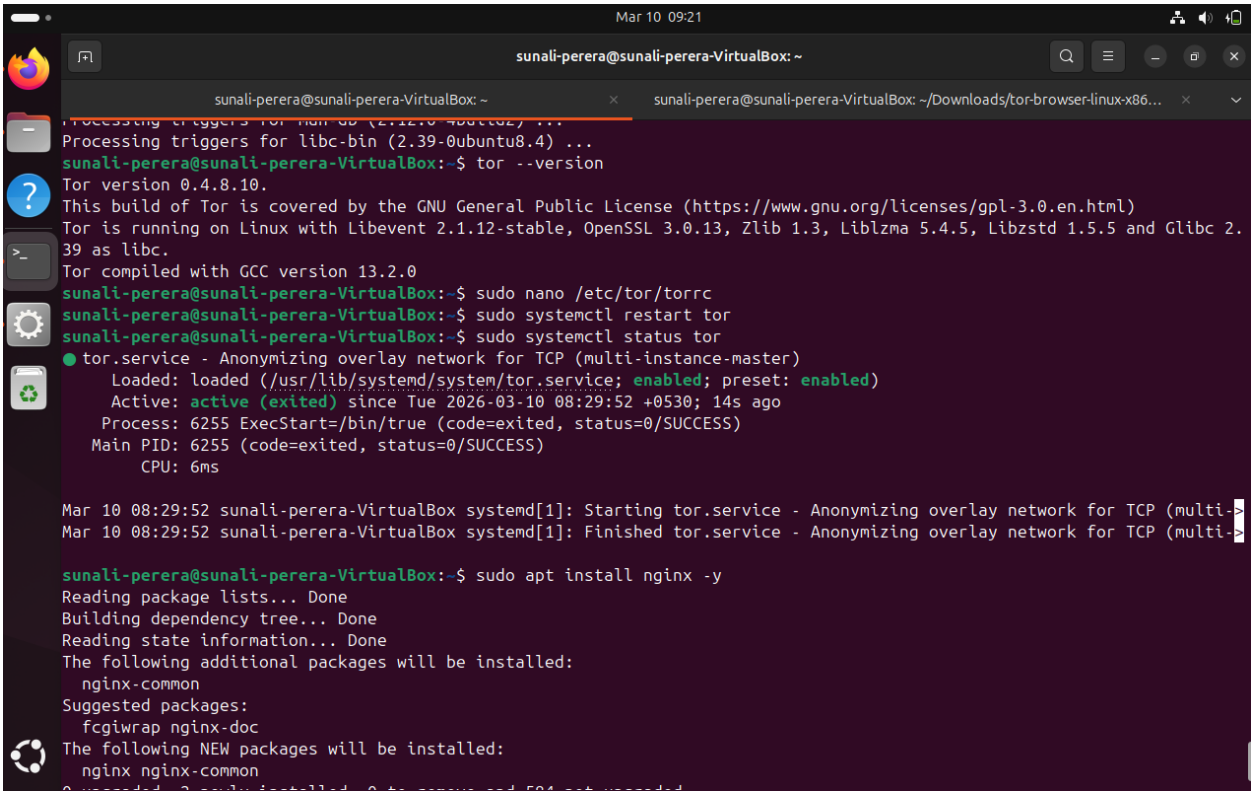
Lab - 14: Setting Up a Tor Hidden Service

23001382



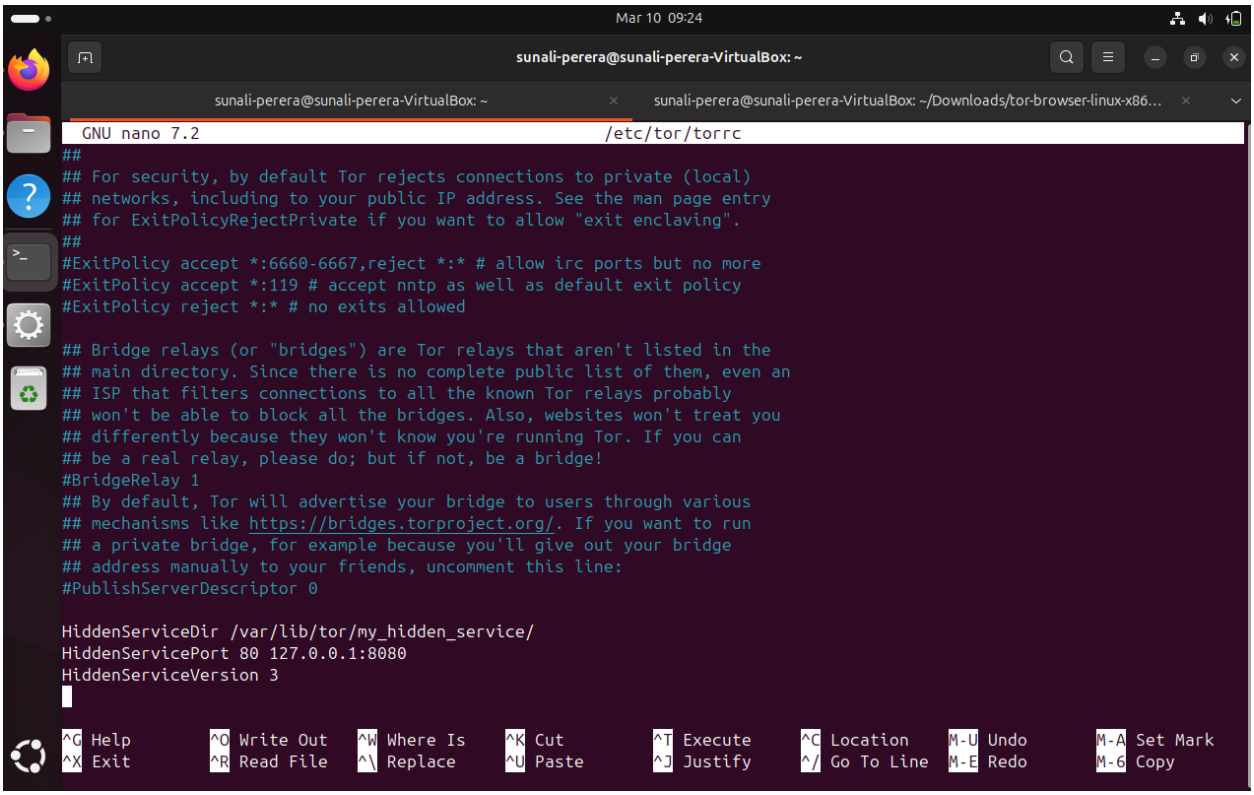
```
sunali-perera@sunali-perera-VirtualBox: ~
sunali-perera@sunali-perera-VirtualBox: ~   sunali-perera@sunali-perera-VirtualBox: ~/Downloads/tor-browser-linux-x86...
E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?
sunali-perera@sunali-perera-VirtualBox:~$ sudo apt install tor -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libevent-2.1-7t64 tor-geoipdb torsocks
Suggested packages:
  mixmaster torbrowser-launcher socat apparmor-utils nyx obfs4proxy
The following NEW packages will be installed:
  libevent-2.1-7t64 tor tor-geoipdb torsocks
0 upgraded, 4 newly installed, 0 to remove and 584 not upgraded.
Need to get 2,833 kB of archives.
After this operation, 15.4 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libevent-2.1-7t64 amd64 2.1.12-stable-9ubuntu2 [145 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 tor amd64 0.4.8.10-1build2 [1,367 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 torsocks amd64 2.4.0-1 [69.9 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 tor-geoipdb all 0.4.8.10-1build2 [1,252 kB]
Fetched 2,833 kB in 4s (678 kB/s)
Selecting previously unselected package libevent-2.1-7t64:amd64.
(Reading database ... 201227 files and directories currently installed.)
Preparing to unpack .../libevent-2.1-7t64_2.1.12-stable-9ubuntu2_amd64.deb ...
Unpacking libevent-2.1-7t64:amd64 (2.1.12-stable-9ubuntu2) ...
Selecting previously unselected package tor.
Preparing to unpack .../tor_0.4.8.10-1build2_amd64.deb ...
Unpacking tor (0.4.8.10-1build2) ...
Selecting previously unselected package torsocks.
Preparing to unpack .../torsocks_2.4.0-1_amd64.deb ...
Unpacking torsocks (2.4.0-1) ...
Selecting previously unselected package tor-geoipdb.
```

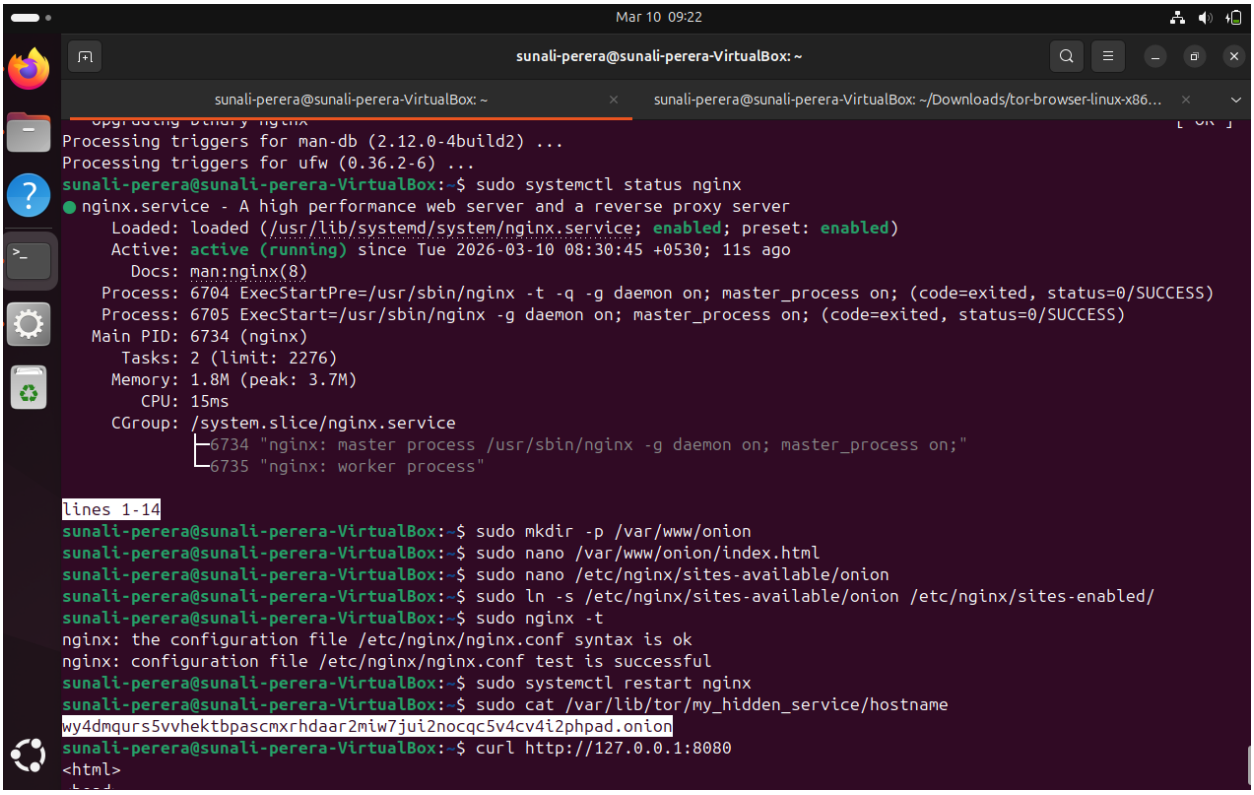


```
Mar 10 09:21
sunali-perera@sunali-perera-VirtualBox: ~
sunali-perera@sunali-perera-VirtualBox: ~   sunali-perera@sunali-perera-VirtualBox: ~/Downloads/tor-browser-linux-x86...
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
sunali-perera@sunali-perera-VirtualBox:~$ tor --version
Tor version 0.4.8.10.
This build of Tor is covered by the GNU General Public License (https://www.gnu.org/licenses/gpl-3.0.en.html)
Tor is running on Linux with Libevent 2.1.12-stable, OpenSSL 3.0.13, Zlib 1.3, Liblzma 5.4.5, Libzstd 1.5.5 and Glibc 2.
39 as libc.
Tor compiled with GCC version 13.2.0
sunali-perera@sunali-perera-VirtualBox:~$ sudo nano /etc/tor/torrc
sunali-perera@sunali-perera-VirtualBox:~$ sudo systemctl restart tor
sunali-perera@sunali-perera-VirtualBox:~$ sudo systemctl status tor
● tor.service - Anonymizing overlay network for TCP (multi-instance-master)
     Loaded: loaded (/usr/lib/systemd/system/tor.service; enabled; preset: enabled)
     Active: active (exited) since Tue 2026-03-10 08:29:52 +0530; 14s ago
    Process: 6255 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 6255 (code=exited, status=0/SUCCESS)
        CPU: 6ms

Mar 10 08:29:52 sunali-perera-VirtualBox systemd[1]: Starting tor.service - Anonymizing overlay network for TCP (multi-
Mar 10 08:29:52 sunali-perera-VirtualBox systemd[1]: Finished tor.service - Anonymizing overlay network for TCP (multi-

sunali-perera@sunali-perera-VirtualBox:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc
The following NEW packages will be installed:
  nginx nginx-common
```


Mar 10 09:24
sunali-perera@sunali-perera-VirtualBox: ~
sunali-perera@sunali-perera-VirtualBox: ~
sunali-perera@sunali-perera-VirtualBox: ~/Downloads/tor-browser-linux-x86...
GNU nano 7.2 /etc/tor/torrc
##
For security, by default Tor rejects connections to private (local)
networks, including to your public IP address. See the man page entry
for ExitPolicyRejectPrivate if you want to allow "exit enclaving".
##
#ExitPolicy accept *:6660-6667,reject *:* # allow irc ports but no more
#ExitPolicy accept *:119 # accept nntp as well as default exit policy
#ExitPolicy reject *:* # no exits allowed
Bridge relays (or "bridges") are Tor relays that aren't listed in the
main directory. Since there is no complete public list of them, even an
ISP that filters connections to all the known Tor relays probably
won't be able to block all the bridges. Also, websites won't treat you
differently because they won't know you're running Tor. If you can
be a real relay, please do; but if not, be a bridge!
#BridgeRelay 1
By default, Tor will advertise your bridge to users through various
mechanisms like https://bridges.torproject.org/. If you want to run
a private bridge, for example because you'll give out your bridge
address manually to your friends, uncomment this line:
#PublishServerDescriptor 0
HiddenServiceDir /var/lib/tor/my_hidden_service/
HiddenServicePort 80 127.0.0.1:8080
HiddenServiceVersion 3
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy


Mar 10 09:22
sunali-perera@sunali-perera-VirtualBox: ~
sunali-perera@sunali-perera-VirtualBox: ~
sunali-perera@sunali-perera-VirtualBox: ~/Downloads/tor-browser-linux-x86...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
sunali-perera@sunali-perera-VirtualBox:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
Active: active (running) since Tue 2026-03-10 08:30:45 +0530; 11s ago
Docs: man:nginx(8)
Process: 6704 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
Process: 6705 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
Main PID: 6734 (nginx)
Tasks: 2 (limit: 2276)
Memory: 1.8M (peak: 3.7M)
CPU: 15ms
CGroup: /system.slice/nginx.service
├─6734 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
└─6735 "nginx: worker process"
lines 1-14
sunali-perera@sunali-perera-VirtualBox:~$ sudo mkdir -p /var/www/onion
sunali-perera@sunali-perera-VirtualBox:~$ sudo nano /var/www/onion/index.html
sunali-perera@sunali-perera-VirtualBox:~$ sudo nano /etc/nginx/sites-available/onion
sunali-perera@sunali-perera-VirtualBox:~$ sudo ln -s /etc/nginx/sites-available/onion /etc/nginx/sites-enabled/
sunali-perera@sunali-perera-VirtualBox:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
sunali-perera@sunali-perera-VirtualBox:~$ sudo systemctl restart nginx
sunali-perera@sunali-perera-VirtualBox:~$ sudo cat /var/lib/tor/my_hidden_service/hostname
wy4dmqurs5vvhektbpascmxrhdaar2miw7jui2nocqc5v4cv4i2phpad.onion
sunali-perera@sunali-perera-VirtualBox:~$ curl http://127.0.0.1:8080
<html>

```
Mar 10 09:24
sunali-perera@sunali-perera-VirtualBox: ~
sunali-perera@sunali-perera-VirtualBox: ~    sunali-perera@sunali-perera-VirtualBox: ~/Downloads/tor-browser-linux-x86...
  GNU nano 7.2                                  /var/www/onion/index.html
<html>
<head>
<title>My Tor Hidden Service</title>
</head>
<body>
<h1>Hello from my Onion Service!</h1>
<p>This is running inside the Tor network.</p>
</body>
</html>

                                              [ Read 9 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
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sunali-perera@sunali-perera-VirtualBox:~$ sudo ln -s /etc/nginx/sites-available/onion /etc/nginx/sites-enabled/
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sunali-perera@sunali-perera-VirtualBox:~$ sudo cat /var/lib/tor/my_hidden_service/hostname
wy4dmqurs5vvhektbpascmxrhdaar2miw7jui2nocqc5v4cv4i2phpad.onion
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sunali-perera@sunali-perera-VirtualBox:~$
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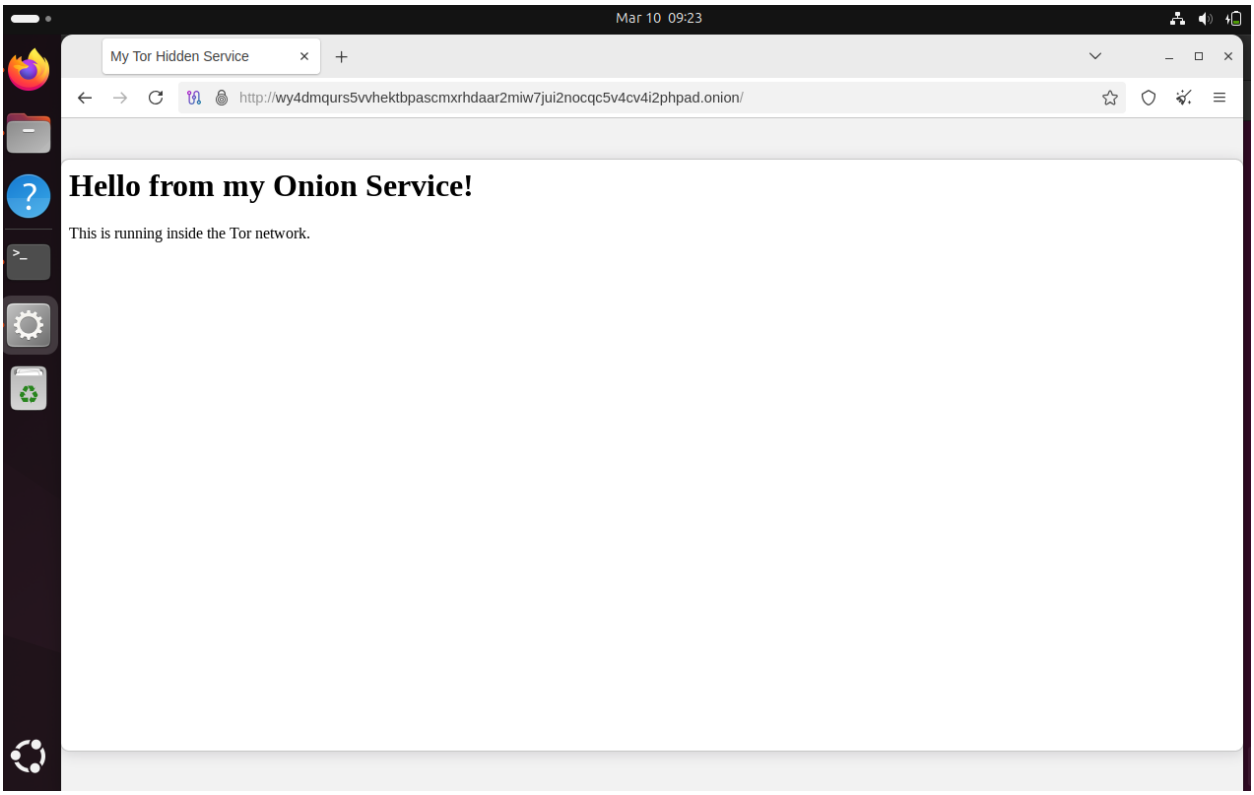
Mar 10 09:23
My Tor Hidden Service
http://wy4dmqurs5vvhektbpascmxrhdaar2miw7jui2nocqc5v4cv4i2phpad.onion/
Hello from my Onion Service!
This is running inside the Tor network.